UNIVERSITY OF LONDON

MSci EXAMINATION 2023

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

CS4860: Advanced Distributed Systems and Communication Networks–

Time Allowed: **TWO hours**

Please answer **ALL** questions.

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1. (a) Which of the following statements are safety conditions, and which are liveness? **Briefly** justify your answer.

   - The Paxos protocol should not allow two different processes to decide on different values.
   - The program should not produce any output in the first hour.
   - The gossip protocol must ensure that eventually a result is collected at the sink from every non-faulty sensor.

   [3 marks]

   (b) Draw a space-time diagram of 3 processes p_0, p_1, and p_2 that contains at least 5 messages. You can choose the sources and destinations of these messages as you wish. Identify 2 consistent cuts and 2 inconsistent cuts in your space-time diagram. [8 marks]

   (c) Compare and contrast the advantages and disadvantages of logical clocks and vector clocks. [4 marks]

   (d) Let $v(e)$ denote the vector clock of event e, and $t(e)$ denote the Lamport logical clock of event e. If two vector clocks for events a, b are incomparable we write $v(a) <> v(b)$.

   Which of the following statements is/are true? **Briefly** justify your answer.

   i. $t(a) = t(b)$ implies that $a = b$

   ii. $v(a) < v(b)$ and $t(b) < t(c)$ implies that $v(a) < v(c)$

   [6 marks]

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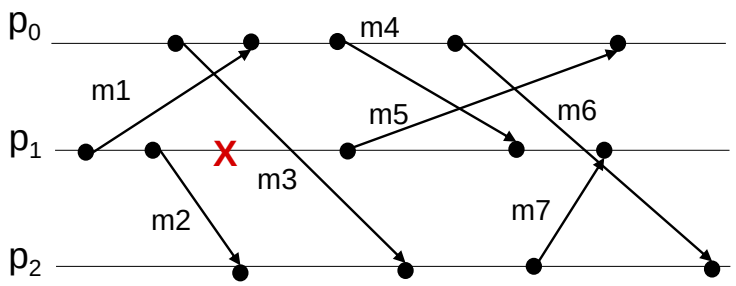


FIGURE 1: SPACE-TIME DIAGRAM WITH DISTRIBUTED SNAPSHOT INITIATED AT TIME X

(e) In the *Distributed Snapshot* problem, one or more processes initiates a snapshot in order to find a consistent cut that includes the state of at least one of the initiators at the time of initiation. Consider the execution of a distributed system shown in Figure 1. If process p_1 initiates the distributed snapshot algorithm discussed in the lectures (extended to handle channel states) at the time indicated with an 'X', is it possible for all the channels to be recorded as empty? **Briefly** explain your answer.

[4 marks]

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2. (a) Consider the 2PC commit protocol and a scenario where there is a coordinator C and three participants P1, P2, and P3. Suppose we have the following sequence of events:

```
1 C sends Prepare Transaction T1 to P1,P2,P3
2 P1 sends Prepared to C
3 P3 sends Prepared to C
4 P1 crashes
5 P2 sends Abort to C
6 C sends Abort Transaction T1 to P2
7 P1 comes back online
8 C crashes
9 P2 crashes
```

What can P1 and P3 do to recover from this situation? [5 marks]

(b) Consider a synchronous network with n processors where up to f processors may fail by crashing and $f \leq n - 2$. The following pseudo-code describes an algorithm that achieves consensus in such a setting:

```
// Code for processor p_i, 0 <= i <= n-1
Initially V = {x}  // V contains p_i's input
In round k, 1 <= k <= f + 1:
   send {v in V : p_i has not already sent v} to all processors.
   receive S_j from p_j, 0 <= j <= n-1, j != i
     V := V U (U {j=0..n-1} S_j)
     if k = f + 1 then y := min(V)
```

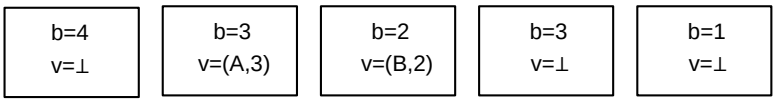
i. Give the definition of the three properties of the consensus problem when processes can fail by crashing. [3 marks]

ii. Consider a slightly modified version of the algorithm above that terminates after only `f` rounds instead of `f+1` rounds. Which of the three consensus properties might be violated? To justify your reasoning, give an example execution (e.g. using a space-time diagram) that violates the property for $f = 1$ and n = 4. [5 marks]

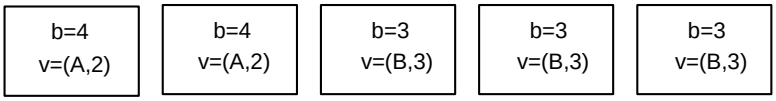
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(c) Consider a Paxos deployment of 5 nodes $p_1 \ldots p_5$, all working on the same consensus instance. Let b denote the last ballot number *promised* by a node, and $v = (v_{acc}, v^b_{acc})$ denote the last value *accepted* by a node (v_{acc}) and the ballot it was accepted (v^b_{acc}). $v = \perp$ denotes no value was accepted yet.

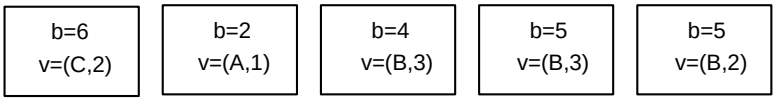
i. Given the following states for nodes $p_1 \ldots p_5$, what value or values can p_1, the leader propose in the next ballot (ballot 5)? Justify your answer. [2 marks]



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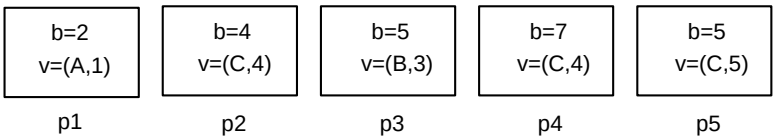
iii. Is the following a possible state in a valid Paxos run? Why or why not? [4 marks]



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iv. Is the following a possible state in a valid Paxos run? Why or why not? [4 marks]



(d) Consider a *crash-recovery* failure model that extends the crash-failure model discussed in the lectures. In the crash-recovery model, processes are equipped with stable storage (e.g. a hard disk), allowing them to persist parts of their state. Any process may fail and then recover, but after a certain amount of time at most f processes will ever fail again. Could the Paxos algorithm be adapted to ensure consensus in the crash-recovery model, and if so what state should leaders and followers store persistently? [5 marks]

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3. (a) Describe three ordering properties that can be required when delivering broadcast messages. [6 marks]

   (b) Derrit.com, a social media forum website, allows its users to create new discussion subforums on which to post and view comments.

   To handle the large number of user requests, Derrit.com is implemented as a decentralized application such that each client's web browser participates in a distributed broadcast algorithm to send and receive messages to other clients whenever a new subforum is created or a comment is posted.

   To avoid confusing users, Derrit.com wishes to ensure that if a user makes a comment c_2 in response to a comment c_1 on the same forum, no user sees a forum with c_2 appearing before c_1. Similarly, if a user creates a new forum f, no comment c_f on that forum should be delivered to another user before the message indicating the new forum has been created. Give the definition of a suitable broadcast ordering guarantee to enforce these properties. Justify your reasoning. [6 marks]

   (c) Consider the sequence of read and write operations shown in Figure 2. State whether the operations shown conform to the sequential specification for read/write registers, where X and Y are two registers. You may assume the start of an interval indicates the invocation time of a read/write operation and the end of the interval represents the time of the response. **Briefly** justify your answer. [4 marks]

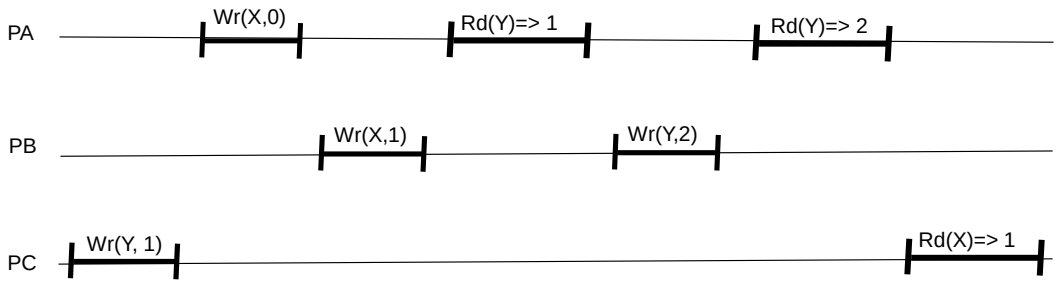


FIGURE 2: SEQUENCE OF READ/WRITE OPERATIONS

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(d) Consider the shared object execution sequence in Figure 3 for two processes p_A and p_B. Assume all writes are to the same shared object X and both processes read continuously. State whether the following operation orderings observed by p_A and p_B are linearizations of the execution sequence, sequentially consistent, both or neither. **Briefly** justify your answers.

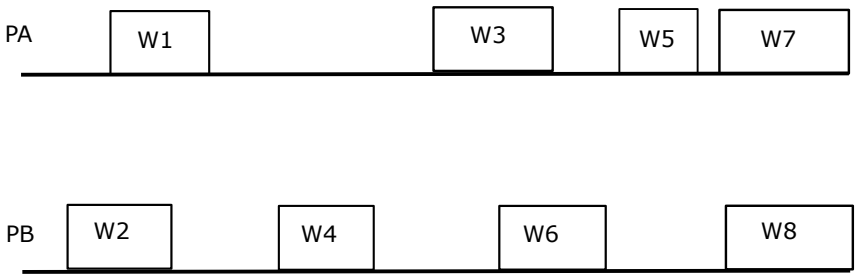


FIGURE 3: SHARED OBJECT EXECUTION SEQUENCE

i. p_A: W1,W2,W4,W3,W6,W5,W7,W8, and
   p_B: W2,W1,W4,W3,W6,W5,W7,W8
ii. p_A and p_B: W2,W1,W4,W3,W6,W5,W8,W7
iii. p_A and p_B: W2,W1,W6,W4,W3,W5,W8,W7
iv. p_A and p_B: W1,W3,W5,W2,W4,W7,W6,W8

[8 marks]

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4. (a) Explain the purpose of the following fields in a Bitcoin block header.
   i. Difficulty target [2 marks]
   ii. Nonce [2 marks]
   iii. Previous block hash [2 marks]

   (b) Explain the purpose of consistent hashing in the context of the Chord peer-to-peer system. [4 marks]

   (c) In the context of storage in peer-to-peer networks, suppose that you are given an ID-cycle of size $32 = 2^5$, where the vertices are numbered in the order $0, \ldots, 31$. The following nodes are connected according to the edges of the ID cycle: p_0 with ID 2, p_1 with ID 8, p_2 with ID 22, and p_3 with ID 30.
   i. Create a diagram of the ID-cycle above showing clearly the position of each node on the ring. [4 marks]
   ii. Assuming appropriate shortcut connections have been set up in order to establish a Chord topology, list the hops taken for a $lookup_i(k)$ operation for key $k = 1$ starting from process $p_i = p_1$. [7 marks]

END

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